

Practice Registering Medical Volumes

The example you just saw is provided for you in the course files. Navigate to the Module 2 folder and open the **RegisteringMedicalVolumes.mlx** file to try the example yourself.

We recommend working with the code in MATLAB to experiment with the code and explore the results in more detail. A static version of the code is available below. The code will not look the same here as interactive tools are used in MATLAB.

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Registering Medical Volumes

Registering medical volumes is a common task for both clinical applications and research studies. This script provides a workflow you can use for registering medical volumes. If you need to register medical images, continue through the course to learn how to use the Registration Estimator App.

Import the Data

Use the controls below to select the T1_3T_2018 and the T1_7T_2020 folders located inside the MRI Images. They are inside the Data folder provided with the course files.

Run this section after selecting for image folders.

Files

T1_3T_2018

T1_7T_2020

```
vol3TFolder = "";
vol7TFolder = "";
```

Import the data using the `medicalVolume` function.

```
vol3T = medicalVolume(vol3TFolder)
vol7T = medicalVolume(vol7TFolder)
```

Examine the metadata

Look at the metadata properties above for each volume. The resolution and plane mapping are different. Registration works best when the geometries of the volumes are the same.

Visualize the volumes

Notice that the 7 Tesla volume covers a smaller portion of the body while the 3 Tesla volume includes the neck and shoulders.

```
orthosliceviewer(vol3T)
orthosliceviewer(vol7T)
```

Resample One Volume

It is usually necessary to ensure your volumes have the same plane mapping and voxel spacing before registration. If they do not, use the `resample` function to resample the voxels to the same geometry. Here, we resample the 3 Tesla volume to the geometry of the 7 Tesla. Often the choice of which volume to resample doesn't matter. In this example, resampling the 3 Tesla volume helps because it removes the neck and shoulders so that both volumes will only include the head.

```
resampled3T = resample(vol3T, vol7T.volumeGeometry)
```

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